

Adopted Levels, Gammas

$Q(\beta^-)=10470$ 40; $S(n)=2470$ 40; $S(p)=18680$ 40; $Q(\alpha)=-13690$ 40 2021Wa16

$S(2n)=10020$ 40, $S(2p)=33930$ 40, $Q(\beta^-n)=2090$ 40 (2021Wa16).

Isotope discovery (2012Th10): $^{232}\text{Th}(^{40}\text{Ar},X)$ at Dubna (1971Ar32).

^{35}Si production:

2015Mo17: $^9\text{Be}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=95$ MeV/nucleon at RIKEN. Measured angular distributions and transverse momentum distributions of fragments. Deduced formulation for the width of transverse momentum distribution as a function of fragment velocity.

2012Kw02: $E(^{40}\text{Ar})=140$ MeV/nucleon impinged on ^9Be , natural nickel, and ^{181}Ta targets at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

2012Zh06: $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=57$ MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.

2007No13: $^9\text{Be}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=100$ MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.

2006Ro34: $^2\text{H}(^{42}\text{S},X)$ at $E(^{42}\text{S})=99.8$ MeV/nucleon at NSCL. Measured production cross sections.

1997Fo01: $^{208}\text{Pb}(^{37}\text{Cl},X)$ at $E(^{37}\text{Cl})=230$ MeV at Legnaro. Measured yields.

^{35}Si decay measurements:

1986Du07, 1986HuZW, 1987DuZU, 1988DuZS, 1988DuZT: $^9\text{Be}(^{40}\text{Ar},X)$ at GANIL. Measured $T_{1/2}$ and β^- -delayed γ rays.

2007Ne14: Polarized ^{35}Si from $^9\text{Be}(^{36}\text{S},X)$ 1n pickup at GANIL. ^{35}Si g.s. magnetic moment and g -factor using β -NMR.

^{35}Si radius measurements:

2006Kh08: ^{35}Si produced by $^{181}\text{Ta}(^{48}\text{Ca},X)$ fragmentation at $E(^{48}\text{Ca})=60.3$ MeV/nucleon at GANIL. Measured energy-integrated reaction cross sections at 30-65 MeV/nucleon using a silicon telescope as both active target and detector. Deduced reduced strong absorption radii, isospin dependence, and possible halo structure or large deformation.

1999Ai02: $\text{Si}(^{35}\text{Si},X)$ at NSCL. Measured energy-integrated reaction cross sections at $E=38$ -80 MeV/ nucleon. Deduced strong absorption radii.

^{35}Si mass measurements: 1986Fi06, 1986Sm05, 1984Ma49.

Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels, J^π , etc.): 2011Ka03, 2009No01, 2008Wi11, 2007Ch82, 2004Kh16, 1999Du05, 1994Mo37, 1994Po05, 1987Wa10, 1986Wo02.

 ^{35}Si LevelsCross Reference (XREF) Flags

A	^{35}Al β^- decay (38.1 ms)	D	$^2\text{H}(^{34}\text{Si}, ^{35}\text{Si}\gamma)$
B	^{36}Al β^-n decay (12.0 ms)	E	$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$
C	^{37}Al β^-2n decay (11.4 ms)	F	$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{Si}\gamma)$

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	XREF	Comments
0	$(7/2)^-$	0.78 s 12	ABCDEF	$\% \beta^- = 100$; $\% \beta^- n < 5$ (1995ReZZ, 2008ReZZ) $\mu = (-)1.639$ 4 (2007Ne14, 2019StZV) μ : From original $(-)1.638$ 4 using β -NMR (2007Ne14) with corrections by 2019StZV. J^π : $L=3$ from 0^+ in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$ and $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si})$ and $\nu f_{7/2}$ configuration from shell model. $T_{1/2}$: From $\beta\gamma(t)$ (1988DuZS, 1988DuZT). Other: 0.87 s 17 by the same group (1986Du07). Reduced strong absorption radius $r_0^2 = 1.261$ fm ² 35 from the energy-integrated σ of $\text{Si}(^{35}\text{Si}, X)$ (2006Kh08) and $r_0^2 = 1.258$ fm ² 92 from the energy-integrated σ of $\text{Si}(^{35}\text{Si}, X)$ (1999Ai02).
909.95 23	$(3/2)^-$	55 ps 14	ABCDEF	J^π : $L=1$ from 0^+ in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$ and $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si})$ and $\nu p_{3/2}$ configuration from shell model. $T_{1/2}$: From analysis of broadened line shapes in $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$.

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Adopted Levels, Gammas (continued) ^{35}Si Levels (continued)

E(level) [†]	J ^π	T _{1/2}	XREF	Comments
973.88 18	(3/2 ⁺)	5.9 ns 6	A EF	J ^π : $\nu d_{3/2}$ configuration from shell model and 715 γ from 1689, 1/2 ⁺ . T _{1/2} : From $\beta\gamma(t)$ in ^{35}Al β^- decay.
1689.4 28	1/2 ⁺		E	J ^π : L=0 from 0 ⁺ in $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si})$.
1970 6			E	
2044 5	(1/2) ⁻		DE	J ^π : L=1 from 0 ⁺ in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$ and $\nu p_{1/2}$ configuration from shell model.
2168.2 4	(5/2 ⁺)		A E	J ^π : From shell model and its possible isobaric analog state in ^{35}P with L=2 from 0 ⁺ in $^1\text{H}(^{34}\text{Si}, p)$ from R-matrix analysis in 2012Im01. L=2,3 from 0 ⁺ in $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si})$.
2275 6			E	
2377 7			E	
3140	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	
3450	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	
3611? 8			E	XREF: E(?) Level exists assuming that the 3611 γ feeds the ground state of ^{35}Si (2014St18).
3770	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	
5190	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	
≈5500	(5/2) ⁻		D	E(level): A broad level at ≈5500 deduced from E _p in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$. J ^π : L=3 from 0 ⁺ in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$ and $\nu f_{5/2}$ configuration from shell model.
5760	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	
6330	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	
7360	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	
7690	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	

[†] From a least-squares fit to γ -ray energies for levels connected with γ transitions; from ^{35}Al β^- -delayed neutron decays for other levels, unless otherwise noted.

[‡] Possible allowed β^- feeding from (5/2)⁺ parent in ^{35}Al .

 $\gamma(^{35}\text{Si})$

E _i (level)	J ^π _i	E _{γ} [†]	I _{γ} [†]	E _f	J ^π _f	Mult.	$\alpha^{\#}$	Comments
909.95	(3/2) ⁻	910.11 30	100	0	(7/2) ⁻	[E2]	4.13×10 ⁻⁵ 6	B(E2)(W.u.)=2.4 +8-5 E _{γ} : others: 910 3 from $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si}\gamma)$, 908 4 from $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$, and 910 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{Si}\gamma)$.
973.88	(3/2 ⁺)	64.1 3 973.78 20	100 11.8 24	909.95 0	(3/2) ⁻ (7/2) ⁻	[E1] [M2]	0.0368 8 5.05×10 ⁻⁵ 7	B(E1)(W.u.)=3.52×10 ⁻⁴ +41-34 B(M2)(W.u.)=0.057 +13-12 E _{γ} : other: 974 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{Si}\gamma)$.
1689.4	1/2 ⁺	715 [‡] 4 780 [‡] 4	14.6 [‡] 16 100 [‡] 8	973.88 909.95	(3/2 ⁺) (3/2) ⁻			
1970		1970 [‡] 6	100	0	(7/2) ⁻			
2044	(1/2) ⁻	1134 [‡] 5	100	909.95	(3/2) ⁻			E _{γ} : Other: 1134 6 from $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si}\gamma)$.
2168.2	(5/2 ⁺)	1194.2 4	35 8	973.88	(3/2 ⁺)			

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Adopted Levels, Gammas (continued) $\gamma(^{35}\text{Si})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Comments
2168.2	2168.2 $6^\dagger$	100 20	0	$(7/2)^-$	E_γ : Other: 2164 6 from $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$.
2275	2275 $6^\ddagger$	100	0	$(7/2)^-$	
2377	2377 $7^\ddagger$	100	0	$(7/2)^-$	
3611?	3611 $8^\ddagger@$	100	0	$(7/2)^-$	2014St18 observed this surprising γ transition at 3611 keV, which must depopulate a state that is unbound to neutron emission by at least 1.1 MeV, considering $S(n)=2470$ 40 (2021Wa16). 2014St18 proposes two possible explanations involving hindered neutron emission or multistep process.

† From ^{35}Al β^- decay, unless otherwise noted.

‡ From $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$.

Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

@ Placement of transition in the level scheme is uncertain.

Legend

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)

